

Assignment-03A : Data flow modelling

Write Verilog Codes using dataflow model for the following (**Continuous Assignments**)

1. **4-bit ALU using Dataflow Modeling**

Design a 4-bit ALU supporting addition, subtraction, AND, OR, XOR, and NOT operations using only continuous assignments and the conditional operator.

2. **8-bit ALU with Status Flags**

Design an 8-bit ALU that performs ADD, SUB, AND, OR, XOR and generates Zero, Carry, Overflow, and Negative flags using dataflow statements.

3. **4-bit Carry Look-Ahead Adder**

Design a 4-bit carry look-ahead adder using generate and propagate equations. Do not use an always block or + operator.

4. **16-bit Carry Look-Ahead Adder**

Construct a 16-bit carry look-ahead adder using four 4-bit CLA blocks. Each block must be described using dataflow equations.

5. **4-bit Carry Select Adder**

Design a 4-bit carry-select adder using dataflow modeling. Generate two possible sums for carry-in 0 and 1, then select the correct result using ?:.

6. **4 bit carry Save Adder**

Take three operands A,B,C. (calculate sum and carry and in final stage)

7. **8-bit Carry Select Adder**

Design an 8-bit carry-select adder using two 4-bit sections. Implement the complete circuit using dataflow modeling.

8. **4-bit Binary to Gray Code Converter**

Design a 4-bit binary-to-Gray converter using Boolean equations and continuous assignments.

9. **4-bit Gray to Binary Converter**

Design a 4-bit Gray-to-binary converter using only dataflow modeling. Derive the equations for every output bit.

10. **BCD to Excess-3 Converter**

Design a BCD-to-Excess-3 converter using Boolean equations and continuous assignments. Treat invalid BCD inputs as don't-care conditions.

11. **Excess-3 to BCD Converter**

Design an Excess-3-to-BCD converter using dataflow modeling. Handle invalid input combinations appropriately.

12. **4-bit Magnitude Comparator**

Design a 4-bit comparator producing:

$A > B$

$A = B$

$A < B$

Use only continuous assignments.

13. 8-bit Magnitude Comparator

Design an 8-bit magnitude comparator by cascading two 4-bit comparator sections.

Implement the complete logic using dataflow modeling.

14. Priority Encoder with Valid Output

Design an 8-to-3 priority encoder with a valid output. Input D[7:0] has priority from D7 to D0. Use dataflow modeling only.

15. 16-to-4 Priority Encoder

Design a 16-to-4 priority encoder with a valid output using Boolean expressions and conditional operators.

16. 4-bit Even/Odd Parity Generator and Checker

Design a circuit that generates an even parity bit for a 4-bit input and another circuit that detects a parity error.

17. 8-bit Parity Generator and Checker

Design an 8-bit parity generator/checker using dataflow modeling. Generate both even and odd parity and indicate an error condition.

18. 4:1 Multiplexer Using Only Boolean Equations

Design a 4:1 multiplexer using only Boolean expressions and continuous assignments. Do not use case or always.

19. 16:1 Multiplexer Using 4:1 MUX Hierarchy

Construct a 16:1 multiplexer from 4:1 multiplexers. Describe each MUX using dataflow modeling and connect them hierarchically.

20. BCD Adder

Design a 1-digit BCD adder using dataflow modeling. The circuit should add two BCD digits and produce:

- BCD sum
- Decimal carry

Implement the correction logic using Boolean/dataflow expressions.